

ARUN KUMAR A N

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [WhatsApp](#)

PROFILE SUMMARY

Full Stack Developer with hands-on experience in enterprise software development and full-stack application development using **React, TypeScript, NestJS, MySQL, and REST APIs**. Experienced in developing modular backend services, scalable application workflows, asynchronous processing, database-driven applications, and responsive user interfaces. Strong understanding of software architecture, API development, debugging, system-level workflows, authentication, containerized development, and Git-based development practices. Experienced in designing loosely coupled application components with a strong focus on scalability, reliability, maintainability, and clean code.

CORE TECHNICAL SKILLS

Frontend: HTML, CSS, JavaScript, React.js, TypeScript, TanStack Query v5, React Hook Form, Zod, Tailwind CSS

Backend & APIs: Node.js, NestJS, RESTful APIs, TypeORM, JWT Authentication

Databases & Queues: MySQL 8, Redis (Caching & Bull Queues), Relational Database Design

Storage & Cloud: MinIO Object Storage

Testing & Tools: Postman, Swagger, GitHub

PROFESSIONAL WORK EXPERIENCE

Inventech Info Solutions Pvt. Ltd.

Bengaluru, India

Full Stack Developer Intern

Dec 2025 – Present

- Developed full-stack web applications using **React, TypeScript, NestJS, REST APIs, and MySQL**.
- Designed modular backend services and implemented RESTful APIs for application workflows and frontend integration.
- Worked on database design, API integration, authentication, validation, debugging, and application performance improvements.
- Developed reusable React components and responsive user interfaces while maintaining structured frontend-backend communication.
- Applied Git-based development workflows, API testing, debugging, and clean coding practices throughout application development.

ENTERPRISE PLATFORMS ENGINEERED AT INVENTECH

Layout Master – Enterprise Layout Planning & Administration System

- Designed Database Architecture in **MySQL & TypeORM** creating 5 Master Tables: **Persons (Employees), Posts (Designations), Roles, Permissions (Create, View, Update, Delete), and Menus (Navigation items)**.
- Architected 4 Mapping Tables to connect the system: **Role-to-Post Mapping, Post-to-Person Mapping** (with start/end date tenure), **Menu-to-Role Mapping** (sidebar menus), and **Permission-to-Role Mapping** (action authorization).
- Developed **NestJS REST APIs** using Controllers, DTOs with validation (**class-validator**), and TypeORM Services to handle CRUD operations and cascading relationships across all mapping tables.
- Engineered **Dynamic Menu & Action Authorization Engine**, where a user's active Post and Role determine what menus render in the React sidebar and which buttons (Add, Edit, Delete) are enabled or hidden.
- Built Full-Stack Admin Screens in **React, TypeScript, and TanStack Query**, allowing administrators to easily assign employees to posts, map menus to roles, and toggle permissions with real-time UI table updates.
- **Tech Stack:** React, TypeScript, Vite, TanStack Query, Tailwind CSS, NestJS, TypeORM, MySQL, JWT

KEONICS – Karnataka State Electronics Development Corporation (Tenant Space & Procurement ERP)

- **Workflow JSON Schema & Process Graph:** Designed structured JSON process graphs (processmaster.json) capturing taskId, process_id, task_type (STARTTASK, NORMAL, DECISION, ENDTASK), and branching decision_flows (Approve, Clarification, Reject) to govern application lifecycles.
- **Database Schema & Instance Tracking:** Architected TypeORM entities and custom repositories (process_instance, task_instance, task_history) in MySQL to track active workflow pointers, task assignments, and lifecycle states.
- **Role-Based State Machine & Task Mapping:** Implemented a backend state transition engine using a TaskToRole lookup dictionary to dynamically map each workflow task to authorized roles (Estate Officer, Zone Commissioner, Commissioner).
- **Validation & Action Payloads:** Engineered RESTful action endpoints (POST /send-for-approval, PUT /close-task/:id) using NestJS DTOs and class-validator, where officers submit decisions, mandatory audit comments, and verification data.

- **Traceable Audit History:** Logged every transition into a history table containing previous state, new state, acting actor ID (Actor A or Actor B), remarks, and timestamps for compliance auditing.
- **Tech Stack:** React, TypeScript, Ant Design, Redux Toolkit, NestJS, TypeORM, MySQL, AWS S3

IEBA – Workforce & Personal Time Study SaaS Platform

- **Assigned Program & Activity Resolution Engine:** Developed backend APIs to dynamically fetch and filter only the assigned programs and active industrial activities mapped to an employee's department role, ensuring accurate logging.
- **Personal Time Study (PTS) Logging & Validation:** Engineered the core time-entry APIs (/user/timeentry/record, /submit) with strict DTO validation that records activity start/end times and total minutes while preventing overlapping time intervals.
- **Supervisor Review & Approval Pipeline:** Built the supervisor review workflow (/timestudyrecord/:id/review) enabling managers to fetch their reportees' submitted weekly time studies, inspect activity breakdowns, and approve or reject them with feedback.
- **Scheduled Time Study (STS) Configuration:** Architected the scheduled time study module allowing administrators to create time study schedules, assign specific eligible users, and restrict logging strictly within the scheduled time windows.
- **Backend Services & Frontend API Integration:** Implemented modular NestJS controllers, TypeORM services, and relational MySQL schemas, seamlessly integrating with the React frontend to synchronize employee time logs and supervisor approval queues.
- **Tech Stack:** React, TypeScript, Vite, TanStack Query, Tailwind CSS, NestJS, TypeORM, MySQL, JWT

WorkspHERE – Enterprise Workforce & Attendance Management

- **Leave Request Lifecycle & RESTful APIs:** Designed normalized MySQL schemas (leave_requests, attendance) with TypeORM and engineered core REST APIs (POST /leave/apply, GET /leave/my-history) with DTO validation to record employee leave types, date ranges, and reason descriptions.
- **Manager Approval & Hierarchy Workflow:** Developed manager review endpoints (GET /manager/leaves/pending, PUT /leave/:id/status) enabling team leads to fetch their direct reportees' pending leave requests and approve or reject them.
- **Automated Attendance State Synchronization:** Built automated service logic where approving a leave request immediately updates the daily attendance records for those dates, marking status as ON_LEAVE or HALF_DAY to maintain consistent daily attendance records.
- **Redis-Backed Bull Queue Cron Schedulers:** Configured automated NestJS cron schedulers (Friday 6:00 PM weekend reminders, month-end submission alerts, and daily attendance auditors) that push notification jobs into a Redis-backed Bull queue (mail-queue).
- **Asynchronous Email Reminder Pipeline (Nodemailer):** Engineered a decoupled Bull queue worker processor (MailProcessor) integrating Nodemailer to dispatch personalized HTML reminder and deadline warning emails asynchronously in the background.
- **Tech Stack:** React, TypeScript, Redux Toolkit, Ant Design, NestJS, Bull Queues, Redis, MySQL, TypeORM, Nodemailer

CDRMS – Correct Dimension Report Management System (PWD Land Survey)

- **MinIO Object Storage Integration:** Configured on-premise MinIO object storage in NestJS using the AWS S3 SDK (S3ClientService) with forcePathStyle enabled to store site layouts, map sketches, and legal documents.
- **Multipart File Upload API:** Built the file upload endpoint (POST /object-store/upload) using custom NestJS upload interceptors to handle multipart form uploads and validate file types and sizes.
- **Organized Folder Path Generator:** Created an automated file path generator in ObjectStoreService that organizes uploads into clean folder paths (entityType/entityId/storedFileName) in MinIO.
- **Database File Key & Metadata Tracking:** Designed the MySQL object_store table using TypeORM to store file details including the file UUID, MinIO object key, MIME type, and file size.
- **Secure File Streaming API by ID:** Built a file viewing endpoint (GET /files/:id) that looks up the file key by its UUID, fetches the file stream from MinIO, and pipes the raw bytes straight to the frontend for image display and document download.
- **Tech Stack:** NestJS, TypeScript, MinIO, AWS S3 SDK, MySQL, TypeORM, Multer, REST APIs

CAS / CSA – Civic Amenity Site Allotment & Citizen Registration Portal

- **Case Worker Scrutiny & Verification Pipeline:** Engineered the CA-Site application verification module guiding citizen applications through Case Worker scrutiny, document checks, and eligibility evaluation.
- **Workflow JSON Process Graphs:** Designed structured JSON process graphs defining workflow stages, task types, allowed decision branches (Approve, Clarification, Reject, Forward), and role transitions.
- **Role-Based Task Routing & Inbox Queues:** Implemented action endpoints (updateVerificationDecision, bulkUpdateDecision) that evaluate the acting officer's role and zone to dynamically progress application lifecycles and route pending tasks to the next officer's inbox.
- **Traceable Decision History & Audit Trail:** Built an append-only history tracking engine (caSite-application-history-item) that logs every administrative action with acting officer ID, previous state, new state, mandatory review remarks, and timestamps.
- **Tech Stack:** React, TypeScript, Vite, TanStack Query, Tailwind CSS, NestJS, TypeORM, MySQL, Redis

INDEPENDENT PROJECTS & OPEN-SOURCE RESEARCH

Voice Excel Assistant – AI Spreadsheet Client & Backend API

[Frontend GitHub](#) | [Backend GitHub](#)

- Built a hands-free spreadsheet client leveraging the **OpenAI Whisper API** and **Web Speech API** for voice-driven cell data entry with column-aware data parsing.
- Engineered an auto-saving grid engine with 500ms debounce persistence, row-level locking, and state management.
- Integrated real-time **ExcelJS streaming** for high-speed .xlsx file generation and immediate exports.
- Structured modular NestJS backend APIs for audio transcript ingestion, formatting, and file streaming.
- **Tech Stack:** React, TypeScript, OpenAI Whisper, Web Speech API, NestJS

Smart Healthcare System – AI Symptom Analysis & Scheduling (MCA Project)

[GitHub](#)

- Developed an intelligent telemedicine portal featuring AI-guided symptom checking and automated doctor specialty recommendation.
- Built an appointment booking calendar with real-time slot management, patient medical record vaults, and role-based access control.
- Implemented secure authentication with JWT, protected patient data encryption, and doctor prescription management.
- Designed responsive, user-friendly clinical dashboards for patients, doctors, and hospital administrators.
- **Tech Stack:** React, TypeScript, NestJS, MySQL, TypeORM, Tailwind CSS, JWT

Interactive School & Academy Dashboard (Inventech Project)

[GitHub](#)

- Designed a responsive timetable scheduling interface using **React Big Calendar** and Lucide icons.
- Built comprehensive analytics dashboards in **Recharts** displaying student attendance trends and examination metrics.
- Implemented form validation with React Hook Form and Zod for course and student enrollment records.
- **Tech Stack:** React.js, TypeScript, Recharts, React Big Calendar, Tailwind CSS, Zod

Realtime Chat & Messaging Platform

[GitHub](#)

- Engineered a bidirectional WebSocket event architecture using **Socket.IO** for low-latency live messaging.
- Implemented active online user rosters, typing indicator triggers, and message delivery status updates.
- Structured persistent conversation history storage in MongoDB with indexed queries for instant retrieval.
- **Tech Stack:** Node.js, Express.js, Socket.IO, React, MongoDB

Online Multi-Vehicle Rental System (BCA Final Semester Project)

[GitHub](#)

- Built multi-category vehicle catalog filtering by class, pricing, and availability status.
- Engineered reservation booking workflow with automated rental duration calculations and billing breakdowns.
- Integrated automated PDF invoice generation and administrative fleet tracking panels.
- **Tech Stack:** PHP, MySQL, HTML5, CSS3, JavaScript, FPDF

Developer Portfolio

[GitHub](#) | [Live Demo](#)

- Designed and built a modern, production-ready developer portfolio showcasing projects, technical skills, and work experience.
- Integrated smooth animations, contact form handling via EmailJS, and responsive dark/light theme.
- Deployed on Vercel with optimized asset delivery and continuous deployment from GitHub.
- **Tech Stack:** React.js, TypeScript, Tailwind CSS, TanStack Query, EmailJS, Vercel

Pharmacy Management System

[GitHub](#)

- Built application to manage medical inventory, point-of-sale billing, and supplier records.
- Implemented automated low-stock notifications, expiration date alerts, and sales reporting.
- **Tech Stack:** Java, MySQL, Relational Database Design

API TESTING & TOOLING

Postman: Tested REST APIs for status codes, request/response data, authentication, and error handling using JWT-based requests.

Swagger: Integrated Swagger UI with NestJS to document, test, and verify REST API endpoints.

EDUCATION

Master of Computer Applications (MCA)	2024 – 2025
<i>Maharaja Institute of Technology, Mysore</i>	CGPA: 8.5 / 10.0
Bachelor of Computer Applications (BCA)	2021 – 2023
<i>St. Joseph's First Grade College, Mysore</i>	CGPA: 8.4 / 10.0

LANGUAGES KNOWN

English (Professional), Kannada (Native)